Comparative Analysis of SerbisYO Against Other Applications and or Systems

SerbisYO is a browser-based, community-powered platform built on a modern tech stack. Rather than relying on single-dimension ratings that obscure operational realities, Serbisyo empowers the people across communities to search for nearby services on an interactive map, inspect service metrics, and contribute post-visit reports.

Core Features of Serbisyo (Proposed)

1. Five-Pillar 5-point Rating: Every establishment breaks performance down into five (5) operational aspects on a 5-point scale:
    a. Service Quality
    b. Reliability
    c. Time and Efficiency
    d. Information Accuracy
    e. Accessibility
2. “Busy Hours” Tracking: Visual histograms showing the busy hour window of an establishment
3. Trust Verification
4. Service Comparison
5. Accountability Access

SerbisYO aims to align with **SDG 11** (Sustainable Cities and Communities), **SDG 16** (Peace, Justice and Strong Institutions), **SDG 10** (Reduced Inequalities), **SDG 9** (Industry, Innovation and

Taxonomy of Comparable Applications and or Systems

Because Serbisyo bridges spatial features, real-time crowd monitoring, queue transparency, civic accountability, and accessibility mapping there is no single existing system is able to do the exact feature set. Instead, SerbisYO aims to cross multiple software categories:

1. Map-Based Local Discovery & Crowd Sensing: Global mapping apps estimates venue busyness and road delays by using GPS and active user reports (Google Maps and Waze).
2. Consumer Business Review & Commercial Waitlisting: Local directory and review that pairs consumer rating ith storefront and waitlist tools (Yelp).
3. Enterprise & Public-Sector Queue Management Systems (QMS)
4. Philippine E-Governance & Statutory Frameworks: National digital service portals and anti-red-tape mandates represented by the DICT (eGov) and the statutory Citizen’s Charter under RA 11032.

Profiles of Some Comparable Systems

Google Maps (Popular Times, Wait Times, Visit Duration, & Local Guides)

As a leading geospatial discovery and navigation platform, **Google Maps** features **Popular Times**, **Live Visit Information**, restaurant **Wait Times**, and **Typical Visit Duration** on Google Business Profiles. These metrics are automatically derived from aggregated, anonymized background location telemetry using differential privacy to track indoor smartphone density. Additionally, over 50 million global **Local Guides** contribute 1-to-5-star ratings, photos, schedule edits, and accessibility details.

System Features:

- **Automated Busyness Metrics & Live Tracking:** Weekly visitor volume bar charts and real-time congestion updates.
- **Duration Estimates & Dining Wait Times:** Algorithmic tracking of customer dwell times.
- **Crowdsourced Contributions & Accessibility Mapping:** Incentivized user reviews confirming hours, details, and accessibility across 40 million points of interest.
- **Enterprise Profile Administration:** Merchant tools for verified managers to post hours, contacts, and respond to feedback.

Pros (Strengths):

- Extensive global user base and extensive location database.
- Automated Popular Times generation in high smartphone density areas
- Intuitive, familiar map interface.

Cons (Limitations):

- **Blind to Service-Specific Queues & Processing Times:** Passive GPS density tracks overall phone counts and stay durations, but cannot separate **wait times** from **processing times** or identify specific congested departments.
- **Missing Data for Smaller/Rural Establishments:** High thresholds for location privacy leave many local government offices, health centers, and utility outlets without busyness data.
- **Single-Metric Star Rating Bias:** Overall ratings lack structured details on document accuracy or real-time service availability.
- **Privacy Concerns:** Continuous background tracking of location and movement prompts privacy issues for users opting out.

Comparison to SerbisYO:

SerbisYO adopts a Google Maps-inspired map layout and hourly chart, replacing passive GPS tracking with service-specific community data. Unlike Google Maps' single rating and overall dwell time, SerbisYO separates **Waiting Time** from **Processing Time**, tracks

Service Availability (e.g., consultations vs. lab services), measures Information Accuracy, and displays Community Confidence via explicit contribution counts.

Waze (Crowdsourced Real-Time Incident Reporting & Waze for Cities)

Overview & Core Mechanism:

Waze is a crowdsourced GPS platform with over 200 million users and 500,000 Map Editors making 2 billion edits yearly. Drivers submit real-time reports on traffic, hazards, and closures, which Waze combines with GPS data to reroute users dynamically. It features voice-based Conversational Reporting via Gemini AI. Through Waze for Cities, governments share road closure data for crowdsourced congestion analytics, a setup used in the Philippines for live routing and evacuation updates during the Taal Volcano eruption.

Key Features

- **Active Crowdsourced Incident & Hazard Reporting:** Structured reporting of traffic, hazards, weather, and closures, plus conversational voice reporting.
- **Real-Time Validation & Decay ("Still There?"):** Recency-weighted reports confirmed or dismissed by passing drivers.
- **Predictive Drive Planning ("Leave Later" Graph):** Traffic histograms estimating travel times across departure windows.
- **Volunteer Map Editor Hierarchy & Partner Hub:** Tiered community moderation paired with municipal government dashboards.

Pros (Strengths):

- Proves that Filipino commuters are willing to contribute real-time updates when interfaces are fast, structured, and mutually beneficial.
- Strong trust model using recency decay, peer verification, and community editors.
- Proven public-sector data sharing via Waze for Cities.

Cons (Limitations):

- Limited to road navigation; offers no insights *after* arrival at offices, clinics, or banks.
- Requires continuous internet connection and consumes significant GPS battery power.

Comparison to SerbisYO:

Conceptually, SerbisYO is "Waze for Public and Essential Services." Just as Waze transformed road traffic from an unpredictable daily frustration into a transparent,

community-monitored network, SerbisYO transforms physical queues ("pila") at government offices, clinics, pharmacies, and registrars into shared community intelligence. SerbisYO's trust engine Recency Weighting, Multi-Report Averaging, Anti-Spam Checks, and a Moderator Queue directly adapts Waze's crowdsourced verification model to indoor service environments.

Yelp (Consumer Reviews & Yelp Waitlist / Guest Manager)

Overview & Core Mechanism:
Yelp is a prominent local business directory and crowd-reviewed platform that expanded into live queue visibility through **Yelp Waitlist**. Yelp combines long-form user reviews, photo uploads, and business attribute filtering with commercial waitlist features such as Predictive Wait Time and Notify Me, which alert users when lines start forming at participating restaurants and allow remote queue check-in.

Key Features:

- **Rich Business Profiles & Attribute Filtering:** Search by category, open status, distance, price tier, and community ratings.
- **Yelp Waitlist & Predictive Wait Time:** Displays live wait times and lets diners remotely join a waitlist and track their spot via smartphone SMS/app updates.
- **Business Owner Portal:** Enables merchants to claim pages, update hours, and reply to customer reviews.
- **Automated Recommendation / Anti-Spam Filter:** Algorithmic filtering of suspicious or biased reviews.

Pros (Strengths):

- High consumer engagement for lifestyle, dining, and commercial businesses especially in North America.
- Combines discovery (search and reviews) with real-time wait visibility on participating pages.

Cons (Limitations):

- **Commercial Dining Focus & Paid B2B Gatekeeping:** Yelp's live wait times require restaurants to purchase Yelp Guest Manager SaaS, a tool public institutions, barangay clinics, and local utilities do not use.
- **Unstructured Narrative Reviews:** Users must read subjective text reviews to estimate wait times or requirements instead of viewing structured metrics for Information Accuracy or Processing Time.

- **Minimal Presence in Philippine Provincial Municipalities:** Yelp has negligible adoption for civic, educational, and healthcare services in municipalities like Morong or Teresa, Rizal.

Comparison to SerbisYO:
Unlike Yelp Waitlist which mandates paid merchant software, SerbisYO decouples wait-time visibility from business procurement, enabling any establishment, public office, or clinic to display accurate wait times, processing times, and reliability scores through community contributions. SerbisYO also replaces unstructured text reviews with a guided 4-step workflow (*Service* → *Visit* → *Experience* → *Review*) to record discrete operational metrics.